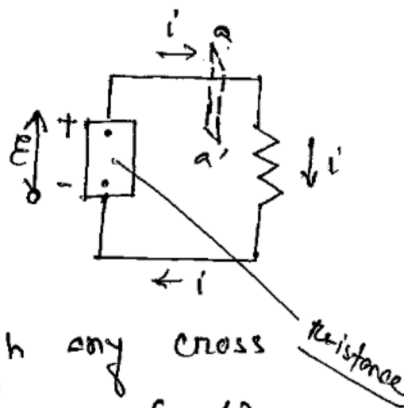


Direct Current Circuits

Chapter 27 - P. H. W 10th Edition.

Single loop circuits If you want to make charge carriers flow through a resistor, you need to establish a potential difference between the ends of the device.

In view of work



A simple electric circuit, in which a device of emf $\mathcal{E}$ does work on the charge carriers and maintains a steady current i in a resistor of resistance R .

In any time interval dt , a charge dq passes through any cross section of this circuit, such as (aa') . This same amount of charge must enter the emf device at its low-potential end and leave at its high-potential end.

□ The device must do an amount of work dW on the charge dq to force it to move in this way.

$$\mathcal{E} = \frac{dW}{dq} \quad (\text{definition of } \mathcal{E}) \quad - \text{work per unit charge.}$$

○ An ideal emf device (internal resistance free)

○ A real emf device (there is an ideal resistance)

Calculating the current in a single-loop circuit.

Energy Method

A charge $dq = i dt$ will have moved through battery $\mathcal{E}$, and the work that the battery will have done on this charge, is

$$dw = \mathcal{E} dq = \mathcal{E} i dt$$

④ From the principle of conservation of energy, the work done by the battery must equal the thermal energy that appears in the resistor.

$$\mathcal{E} i dt = i^2 R dt$$

$$\mathcal{E} = i R$$

$$i = \frac{\mathcal{E}}{R}$$

$P = i^2 R$, tells us that in (26-27) a time interval dt an amount of energy given by i^2

Potential Method

Loop Rule The algebraic sum of the changes in potential encountered in a complete traversal of any loop of a circuit must be zero. (Kirchoff's loop rule) / voltage law.

$$\mathcal{E} - iR = 0 \quad [\text{loop}]$$

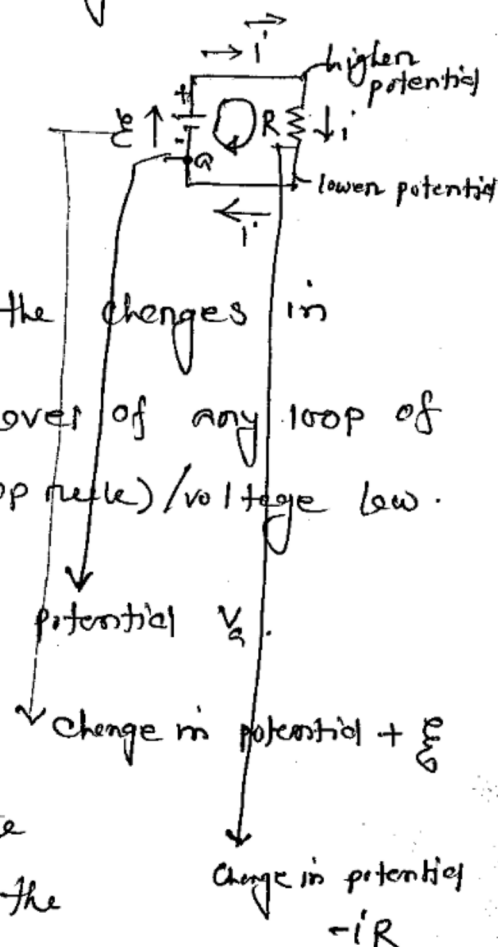
$$i = \frac{\mathcal{E}}{R}$$

If we apply the loop rule to a complete counter-clockwise walk around the circuit, the rule gives us

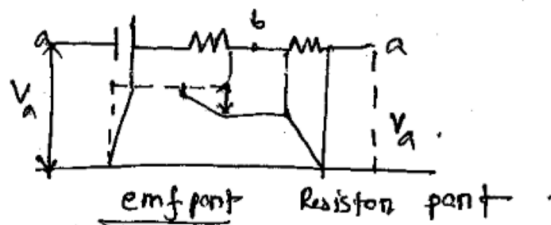
$$-\mathcal{E} + iR = 0$$

$$i = \mathcal{E}/R$$

Thus we can use loop rule any direction of a loop in a circuit.



Internal Resistance

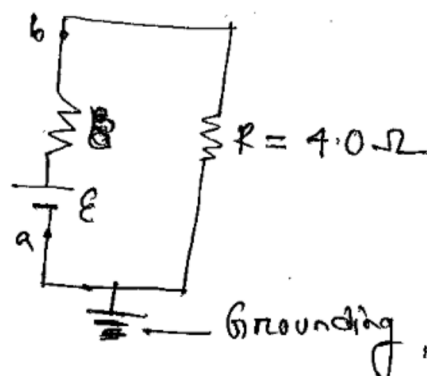


loop rules: $\mathcal{E} - ir - iR = 0$

$$i = \frac{\mathcal{E}}{R + r}$$

□ Draw a circuit where three resistance in circuit, apply loop rules and find current?

Grounding a Circuit



$\equiv$ Grounding a circuit

usually means connecting the circuit to a conducting path to Earth's Surface. Here such a connection means only that the potential is defined to be zero at grounding point in the circuit.

Power, Potential and Emf.

A real emf device has an internal resistance r , it also transfers energy to internal thermal energy via resistive dissipation.

$$P = iv \quad | \quad v = \mathcal{E} - ir$$

$$P = i(\mathcal{E} - ir) = i\mathcal{E} - i^2 r$$

$$P_r = i^2 r \text{ (internal dissipation rate)}$$

$$P_{\text{emf}} = i\mathcal{E} \text{ (power of emf device)} \rightarrow (*)$$

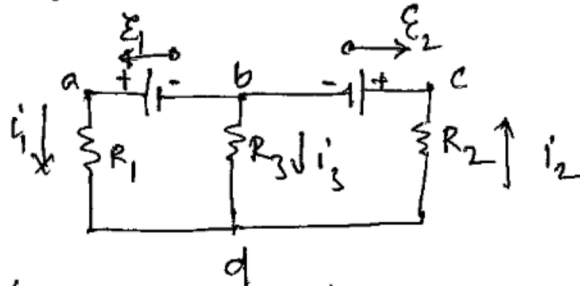
Multiloop Circuits

$$\frac{1}{R_{eq}} = \sum_{j=1}^n \frac{1}{R_j} \quad n \text{ resistance in parallel} \quad (4)$$

Junction Rule: The sum of the currents entering any junction must be equal to the sum of the currents leaving that junction. (Kirchhoff's Current law).

loop rule: conservation of energy.

junction rule: based on the conservation of charge.



$i_1 + i_3 = i_2 \rightarrow$ Total incoming current = Total outgoing current.

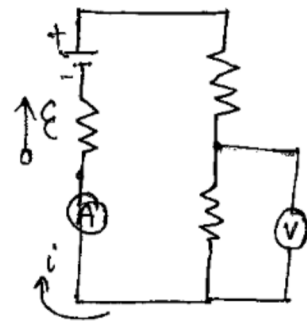
left loop: $\mathcal{E}_1 - i_1 R_1 + i_3 R_3 = 0$

right loop: $-i_3 R_3 - i_2 R_2 - \mathcal{E}_2 = 0$

big loop: $\mathcal{E}_1 - i_1 R_1 - i_2 R_2 - \mathcal{E}_2 = 0$

The Ammeter and the Voltmeter

A single-loop circuit, showing how to connect an ammeter (A) and a voltmeter (V).



RC Circuits time-varying currents

① here we examine charging process.

② How charge $q(t)$ on the capacitor plates
③ the potential difference across the capacitor.

④ $i(t)$ vary with time during the charging process.

Apply loop rule:

$$\mathcal{E} - iR - \frac{q}{C} = 0 \quad \text{--- (I) (Anti-clockwise direction)}$$

potential across the capacitor.

negative as we are moving from low potential to higher potential.

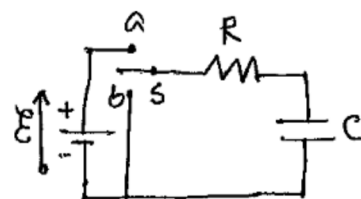
$$i = \frac{dq}{dt} \quad \text{--- (II)}$$

$R \frac{dq}{dt} + \frac{q}{C} = \mathcal{E}$

 (charging equation)

to solve this equation we need to find the function $q(t)$ that satisfies this equation, the capacitor be initially unchanged, $q=0$, and $t=0$.

$$q = C\mathcal{E} (1 - e^{-t/RC}) \quad \text{(charging a capacitor function.)}$$

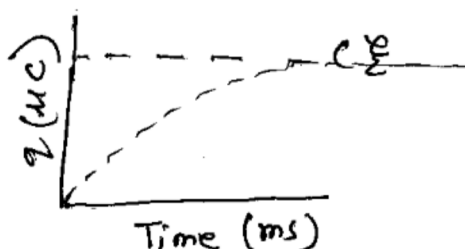


When switch S is closed on a, the capacitor is charged through the resistor. When the switch is afterward closed on b, the capacitor discharges through the resistor.

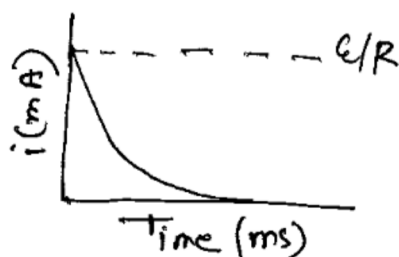
• The derivative of $q(t)$ is the current $i(t)$ charging the capacitor.

$$i = \frac{dq}{dt} = \left(\frac{\epsilon}{R} \right) e^{-t/RC} \quad (\text{charging the capacitor})$$

$$t \rightarrow \infty, \quad q = C\epsilon$$



buildup of charge on the capacitor



decline of the charging current.

▣ A capacitor that is being charged initially acts like ordinary connecting wire relative to the charging current. A long time later, it acts like a broken wire.

$q = CV$, the potential difference $V_c(t)$ across the capacitor during the charging process is

$$V_c = \frac{q}{C} = \epsilon \left(1 - e^{-t/RC} \right) \quad \text{charging capacitor}$$

This tells us that $V_c = 0$ at $t = 0$ and $V_c = \epsilon$ when the capacitor becomes fully charged as $t \rightarrow \infty$.

The time constant

Should be dimensionless, RC is has the unit of time. RC is called the capacitive time constant of the circuit and is represented with the symbol τ .

$$\tau = RC$$

At time $t = \tau = RC \rightarrow q = C\epsilon (1 - e^{-1}) = 0.63 C\epsilon$
 $\tau = 1\mu s$ and $\tau = 100s$ means ?

Discharging a Capacitor

$$R \frac{dq}{dt} + \frac{q}{C} = \mathcal{E} \quad \text{charging solution: } q = C\mathcal{E}(1 - e^{-t/RC})$$

$$R \frac{dq}{dt} + \frac{q}{C} = 0 \quad \text{discharging " } q = q_0 e^{-t/RC}$$

$$\text{Where, } q_0 = CV_0.$$

$$i = \frac{dq}{dt} = -\left(\frac{q_0}{RC}\right) e^{-t/RC} \quad \begin{matrix} \xrightarrow{\tau} \\ \xrightarrow{I_0} \end{matrix}$$

This tells us that the current also decreases exponentially with time, at a rate set by τ .

Perivation

$$R \frac{dq}{dt} + \frac{q}{C} = \mathcal{E}$$

$$\frac{dq}{dt} + \frac{q}{RC} = \frac{\mathcal{E}}{R} \quad \text{general solution}$$

$$q = q_p + k e^{-at} \quad \begin{matrix} \leftarrow \\ a \text{ is the coefficient of } q \end{matrix}$$

$$q_p = C\mathcal{E} \quad \text{particular solution} \quad = \frac{1}{RC}$$

$$q = C\mathcal{E} + k e^{-at}$$

$q_p \rightarrow$ particular solution of the differential eqⁿ.

$k \rightarrow$ constant.

$$q=0, \text{ and } t=0$$

$$0 = C\mathcal{E} + k$$

$$k = -C\mathcal{E}$$

$$q = C\mathcal{E} - C\mathcal{E} e^{-t/RC}$$

$$q = C\mathcal{E} (1 - e^{-t/RC})$$

Solution.